CM2

Probability

Welcome!

- The question of this lecture: when I run a randomized algorithm, how confident can I be that it behaves *close to* its average behavior?
- The honest answer needs *tail bounds* — inequalities such as "the probability of a 10% deviation is at most $e^{-n/100}$". In this lecture we build the tools leading up to them; Chernoff bounds come next time.
- Today we proceed one careful step at a time:
  1. what a random variable is,
  2. expectation and its best property (linearity),
  3. Markov's inequality,
  4. why Markov is often too weak — and a preview of Chernoff bounds, which we prove next lecture.

Notation we will use all the time

Ω	set of all possible outcomes (finite or countable)
$A \subseteq \Omega$	an event; $\Pr[A]$ its probability
$X \colon \Omega \to \mathbb{R}$	a random variable
$X \sim \mathrm{Bin}(n, p)$	binomial: # successes in n independent trials, each with probability p
$\mathbb{E}[X]$	expectation (mean) of X
$\mathbf{1}_A$	indicator of A: equals 1 on A, 0 outside A
w.h.p.	"with high probability" (probability tending to 1 as the input size grows)

Part I: Random variables

- We need a precise language to talk about randomized algorithms.
- Ingredients:
  1. the set Ω of all possible outcomes,
  2. a rule Pr telling us how likely each outcome is,
  3. functions $X : \Omega \to \mathbb{R}$ measuring quantities we care about.

Let us make each ingredient precise, with the example of two dice throughout.

Probability spaces

Definition 1 (Discrete probability space)

A pair $(\Omega, \Pr)$ where

1. Ω is a finite or countably infinite set of *outcomes*;
2. $\Pr$ assigns to every *event* $A \subseteq \Omega$ a number $\Pr[A] \in [0,1]$ with
   1. $\Pr[\Omega] = 1$;
   2. if $A_1, A_2, \ldots$ (countably many) are pairwise disjoint, then $\Pr\left[\bigcup_i A_i\right] = \sum_i \Pr[A_i]$.

For finite Ω it is enough to specify $\Pr[\omega]$ for each outcome ω and then set $\Pr[A] = \sum_{\omega \in A} \Pr[\omega]$.

Example 2 (Two fair dice)

$\Omega = \{1,\ldots,6\} \times \{1,\ldots,6\}$, all 36 pairs equally likely: $\Pr[(i,j)] = \frac{1}{36}$. The event "sum is 7" has probability $\frac{6}{36} = \frac{1}{6}$.

What is a random variable?

Definition 3 (Random variable)

A random variable is a function $X \colon \Omega \to \mathbb{R}$. For a value x:

$$\Pr[X = x] \;:=\; \Pr\big[\{\omega \in \Omega : X(\omega) = x\}\big].$$

Example 4 (Sum of two dice)

$X(i,j) = i + j$. Then

$$\Pr[X = 7] = \Pr[\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}] = \frac{6}{36} = \frac{1}{6}.$$

Key observation: X is just a quantity we compute from the outcome of the random experiment.

The simplest random variables: indicators

Definition 5 (Indicator of an event)

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

- $\mathbf{1}_A$ is 1 exactly when the event A happens — it *counts* A.
- Almost everything later will be a *sum of indicators*:
  - number of heads in n coin tosses,
  - number of samples smaller than a threshold,
  - ...
- So if we understand indicators well, we understand a lot.

Example: binomial random variables

Example 6

Toss n independent fair coins; $X = \#$ heads. Then

$$\Pr[X = j] = \binom{n}{j} 2^{-n},$$

and we write $X \sim \mathrm{Bin}(n, \frac{1}{2})$. (More generally $X \sim \mathrm{Bin}(n, p)$ if each coin lands heads with probability p.)

Example 7 (Indicators again)

With $A_i =$ "coin i lands heads",

$$X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n}.$$

This decomposition will be the workhorse of the whole lecture.

Check yourself

Quick question: We roll two fair dice. What is the probability that the sum is 3?

$$\Pr[X = 3] = \Pr[\{(1,2),(2,1)\}] = \frac{2}{36} = \frac{1}{18}.$$

Part II: Expectation

- A random variable's most basic summary statistic is its *average*: the expectation $\mathbb{E}[X]$.
- Intuition: if you repeat the experiment many times and average the value of X, you converge to $\mathbb{E}[X]$. (This is the law of large numbers.)
- Today we use $\mathbb{E}[X]$ as a *mathematical tool*, so we define it exactly.

Definition of expectation

Definition 8 (Expectation)

For a discrete random variable X on $(\Omega, \Pr)$,

$$\mathbb{E}[X] := \sum_{\omega \in \Omega} X(\omega) \Pr[\omega],$$

where $\Pr[\omega]$ abbreviates $\Pr[\{\omega\}]$, provided the sum converges absolutely (all our examples use finite Ω).

Example 9 (One fair die)

$X \in \{1, \ldots, 6\}$ uniform: every outcome has probability $\frac{1}{6}$, so
$\mathbb{E}[X] = \frac{1}{6}(1 + 2 + \cdots + 6) = \frac{21}{6} = 3.5$.

Notice: the expectation need not be a value that X ever takes!

An equivalent, often more convenient formula

Proposition 10 (Grouping by value)

$$\mathbb{E}[X] = \sum_{x} x \cdot \Pr[X = x],$$

the sum running over all values x that X can take.

Proof.

Split the big sum $\sum_{\omega}$ into groups according to the value of X:

$$\mathbb{E}[X] = \underbrace{\sum_{\omega} X(\omega)\Pr[\omega]}_{\text{one term per outcome}} = \sum_{x} \sum_{\omega:\, X(\omega)=x} X(\omega)\Pr[\omega] = \sum_{x} x \sum_{\omega:\, X(\omega)=x} \Pr[\omega] = \sum_{x} x \Pr[X = x].$$

Every outcome with $X(\omega) = x$ contributes x, and the inner sum equals $\Pr[X = x]$. □

The most important expectation: indicators

Proposition 11 (Expectation of an indicator)

$$\mathbb{E}[\mathbf{1}_A] = \Pr[A].$$

Proof.

Only the outcomes in A contribute ($\mathbf{1}_A = 1$ there, 0 elsewhere):

$$\mathbb{E}[\mathbf{1}_A] = \sum_{\omega} \mathbf{1}_A(\omega) \Pr[\omega] = \sum_{\omega \in A} \Pr[\omega] = \Pr[A].$$

□

Moral: probabilities are expectations of indicators.

This single fact will let us turn counting problems into sums of probabilities.

Two basic properties

Proposition 12

1. *Monotonicity: if $X \leq Y$ pointwise (i.e. $X(\omega) \leq Y(\omega)$ for every ω), then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.*
2. *Scaling: $\mathbb{E}[cX] = c\,\mathbb{E}[X]$ for every constant c.*

Proof.

Both follow from the definition, term by term:

$$\mathbb{E}[cX] = \sum_{\omega} cX(\omega)\Pr[\omega] = c\sum_{\omega} X(\omega)\Pr[\omega] = c\,\mathbb{E}[X],$$

and if $X \leq Y$ then every summand of $\mathbb{E}[X]$ is at most the corresponding summand of $\mathbb{E}[Y]$. □

These will be used constantly, often without saying so.

Part III: Linearity of expectation

- The single most useful fact about expectations:
- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ *always* — even if X and Y are strongly dependent!
- That sounds surprising: means behave nicely even when the variables themselves do not.

Linearity of expectation: statement and proof

Theorem 13 (Linearity of expectation)

For any two random variables X, Y on the same space:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Proof.

The sum $X + Y$ is the function $(X + Y)(\omega) = X(\omega) + Y(\omega)$. Now simply rearrange the sum:

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_{\omega} \big(X(\omega) + Y(\omega)\big) \Pr[\omega] \\ &= \sum_{\omega} X(\omega) \Pr[\omega] + \sum_{\omega} Y(\omega) \Pr[\omega] = \mathbb{E}[X] + \mathbb{E}[Y].\end{aligned}$$

□

Where is independence used? Nowhere! That is exactly the point.

Consequences

Corollary 14

1. *For any $X_1, \ldots, X_n$ (no independence needed):* $\mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i]$.
2. $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ *and* $\mathbb{E}[X - Y] = \mathbb{E}[X] - \mathbb{E}[Y]$.

Proof.

Part 1: apply the theorem repeatedly (induction on n). Part 2: combine the theorem with scaling. □

Remark 15 (Contrast)

$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ *fails in general: independence suffices, but is not necessary. Means always add; products multiply only under extra assumptions (e.g. independence — more generally, zero correlation).*

Example: expectation of a binomial (no heavy computation)

Example 16

$X \sim \mathrm{Bin}(n, p)$: toss n coins, each heads with probability p. Write $X = \mathbf{1}_{A_1} + \cdots + \mathbf{1}_{A_n}$, $A_i =$ "coin i heads".

Using $\mathbb{E}[\mathbf{1}_{A_i}] = \Pr[A_i] = p$ and linearity:

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[\mathbf{1}_{A_i}] = \sum_{i=1}^{n} p = np.$$

- Compare: computing $\sum_j j\binom{n}{j} p^j (1-p)^{n-j}$ directly is painful; this took two lines.
- Linearity did not require the coin tosses to be independent — it holds for any sum of indicators. (Here they are independent, but that is irrelevant for the mean.)

Example: fixed points of a random permutation

Example 17

Shuffle $\{1, \dots, n\}$ uniformly: a random permutation σ. Let $X = \#\{i : \sigma(i) = i\}$ be the number of fixed points.

Write $X = \sum_{i=1}^{n} \mathbf{1}_{\{\sigma(i)=i\}}$. For each fixed i:

$$\Pr[\sigma(i) = i] = \frac{\#\text{permutations with } \sigma(i) = i}{n!} = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Linearity:

$$\mathbb{E}[X] = \sum_{i=1}^{n} \Pr[\sigma(i) = i] = n \cdot \frac{1}{n} = 1.$$

On average exactly one fixed point — for every n.

(Equivalently: on average one guest gets his own hat back.)

Part IV: Markov's inequality

- Expectations are averages. But an average hides the *spread*:
- A variable with $\mathbb{E}[X] = 1$ could always be 1, or could be 10^6 with probability 10^{-6} and 0 otherwise.
- We want upper bounds on *tail probabilities*: $\Pr[X \geq a]$ for large a.
- Question: how much can we say using only the expectation?

Markov's inequality

Theorem 18 (Markov)

If $X \geq 0$ and $a > 0$, then

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Why nonnegativity? Negative values can cancel a large positive tail inside the mean, keeping $\mathbb{E}[X]$ small while $\Pr[X \geq a]$ is large; without $X \geq 0$ the claim is simply false.

Two proofs follow — both instructive, both very short.

Markov's inequality: proof 1 (splitting the sum)

Proof.

Group the formula $\mathbb{E}[X] = \sum_x x \Pr[X = x]$ and drop the small values:

$$\mathbb{E}[X] = \sum_x x \Pr[X = x] \;\geq\; \sum_{x \geq a} x \Pr[X = x].$$

Now every remaining term has $x \geq a$:

$$\sum_{x \geq a} x \Pr[X = x] \;\geq\; \sum_{x \geq a} a \Pr[X = x] = a \sum_{x \geq a} \Pr[X = x] = a \Pr[X \geq a].$$

Together:

$$\mathbb{E}[X] \geq a \Pr[X \geq a] \qquad \Longrightarrow \qquad \Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

□

Markov's inequality: proof 2 (the indicator trick)

Proof.

Compare the two quantities *pointwise*:

$$a \cdot \mathbf{1}_{\{X \geq a\}} \quad \text{vs.} \quad X.$$

- If $X < a$: left side $= 0 \leq X$. ✓
- If $X \geq a$: left side $= a \leq X$. ✓

So $a\mathbf{1}_{\{X \geq a\}} \leq X$ always. Expectations respect inequalities (monotonicity), and $\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \Pr[X \geq a]$:

$$a \Pr[X \geq a] = \mathbb{E}\big[a\mathbf{1}_{\{X \geq a\}}\big] \leq \mathbb{E}[X].$$

□

This "multiply by an indicator, then take expectations" trick will appear again and again.

Consequences and a warning

Corollary 19

If $X \geq 0$ and $\mathbb{E}[X] > 0$, then for every $c > 0$: $\Pr[X \geq c\,\mathbb{E}[X]] \leq \frac{1}{c}$.

Proof.

Take $a = c\,\mathbb{E}[X]$ in Markov's inequality (needs $a > 0$, hence $\mathbb{E}[X] > 0$). □

Example 20 (Markov is weak)

$X \sim \mathrm{Bin}(n, \frac{1}{2})$: $\mathbb{E}[X] = n/2$, so

$$\Pr\left[X \geq \tfrac{3n}{4}\right] \leq \frac{n/2}{3n/4} = \frac{2}{3}.$$

Truth: far smaller — $\leq e^{-n/24}$ (Chernoff, next lecture); Markov sees only the mean, not the independence of the coins.

Markov is actually optimal — no stronger bound is possible

Remark 21 (Tightness)

Let X take value a with probability p and value 0 otherwise. Then

$$\mathbb{E}[X] = ap, \qquad \Pr[X \geq a] = p = \frac{\mathbb{E}[X]}{a}.$$

Markov's inequality is an *equality* for this X:

Using only the mean, nothing stronger than Markov can ever be true.

To beat Markov we must use *more structure* — next lecture: independence and Chernoff bounds.

Next class

- Chernoff bounds: a complete proof.
- Applications to randomized algorithms.